Manufacturing & Video-Analytics Intelligent Report

“I want the data from the mes application for machine utilization and from video analytics I want camera configuration information”

Generated: 2026-10-20 17:49:40
Time filter applied

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Executive Summary

2	12	333	Applied	0
DATABASES QUERIED	TABLES MATCHED	TOTAL RECORDS	TIME FILTER	FAILED QUERIES

Matched keywords: camera, configuration, machine, utilization

Telemetry window: 2026-10-19T12:19:40Z → 2026-10-20T12:19:40Z (bucketed every 5m)

Video Analytics Camera Configuration and Operational Status

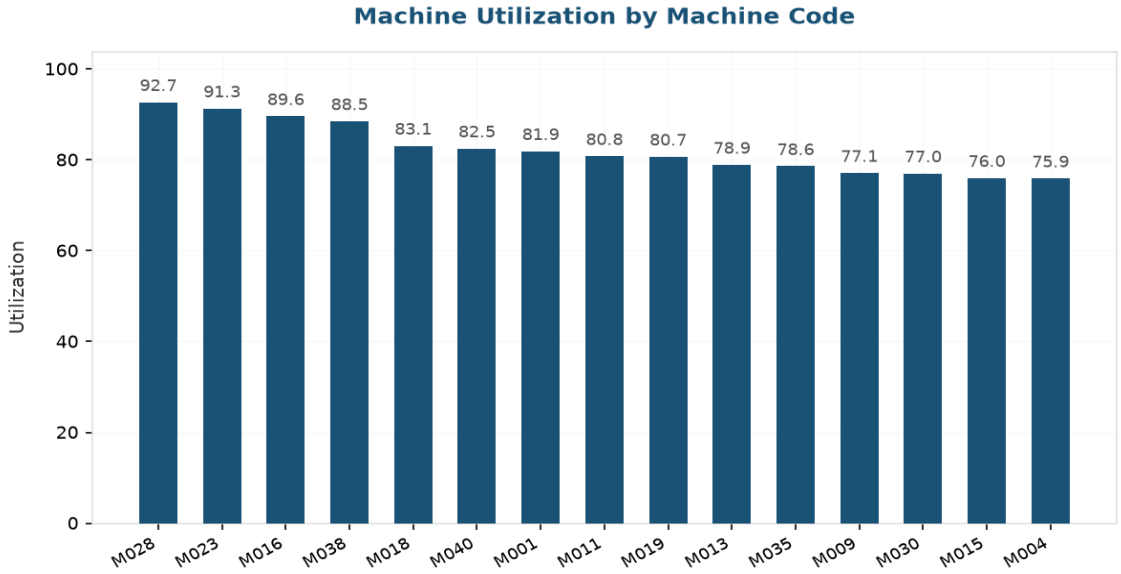
The Video Analytics configuration database contains details for three registered cameras: 'Flarehub' (Camera #1), 'Luxsphere' (Camera #2), and 'Entry Gate' (Camera #6, assigned to Plant 1, Department 1, Location 1). All three cameras are configured under administrator access ('admin') and reflect an 'online' administrative status in the master configuration table. However, examination of the camera_status_logs dataset (50 log entries for camera IDs 1 through 3) reveals a 100% occurrence of 'offline' status records. This operational discrepancy highlights that while administrative configuration parameters are active, real-time video streaming or network connectivity to these cameras is currently broken, representing a critical monitoring gap.

Data preview — cameras (showing 3 of 3 points/rows)

camera_number	department_id	location_id	name	plant_id	status	user_id
1	None	None	Flarehub	None	online	admin
2	None	None	Luxsphere	None	online	admin
6	1	1	Entry Gate	1	online	admin

MES Fleet Machine Utilization Overview

The MES MachineMaster database contains master parameters and performance statistics for 40 manufacturing assets spanning five equipment types: Injection Molding, Thermoforming, Assembly, Extrusion, and Blow Molding. Asset utilization across the fleet averages 68.61%, varying from a minimum of 43.40% up to a high of 92.70%. Machine total capacities average 541.73 units (ranging from 120.46 to 983.19 units) with an average hourly output capability of 216.60 units/hr (min: 52.82, max: 396.75). All assets are currently tied to Customer ID 1, indicating single-tenant operational deployment.

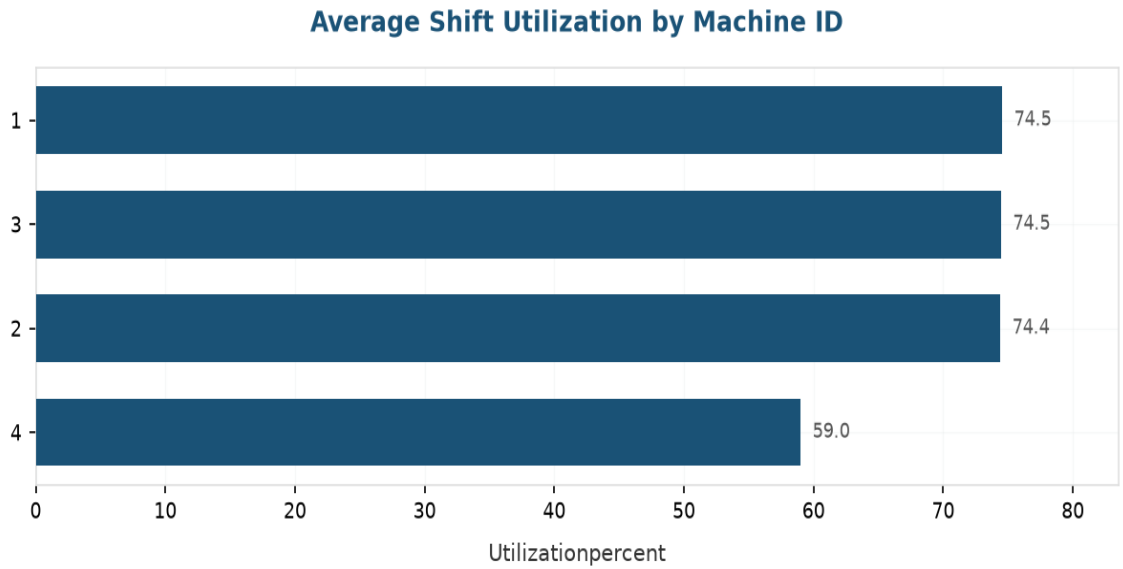


Data preview — MachineMaster (showing 10 of 40 points/rows)

Utilization	MachineCode	MachineId	MachineName	MachineType	Capacity	CapacityPerHour	CustomerId
81.90	M001	1	Injection Molding Line 1	Injection Molding	772.96	178.64	1
45.00	M002	2	Thermoforming Line 2	Thermoforming	554.94	77.34	1
58.90	M003	3	Thermoforming Line 3	Thermoforming	826.22	229.64	1
75.90	M004	4	Thermoforming Line 4	Thermoforming	681.29	150.48	1
59.00	M005	5	Assembly Line 5	Assembly	120.46	235.09	1
69.70	M006	6	Injection Molding Line 6	Injection Molding	190.81	333.32	1
43.40	M007	7	Extrusion Line 7	Extrusion	705.28	335.53	1
56.50	M008	8	Thermoforming Line 8	Thermoforming	572.02	66.41	1
77.10	M009	9	Assembly Line 9	Assembly	283.00	388.73	1
70.40	M010	10	Blow Molding Line 10	Blow Molding	943.44	217.79	1

Shift-Level Capacity and Utilization Analysis

Granular shift performance from the CapacityAnalysis dataset evaluates 50 operational records across Machine IDs 1 through 4 across four active shifts (Shift IDs 1 to 4). Average shift utilization across these primary assets stands at 72.92%, outperforming the general fleet average of 68.61%. However, performance varies widely, ranging from a low shift utilization of 37.50% to maximum capacity utilization of 100.00%. These variance bounds indicate significant shift-to-shift workload fluctuations or localized downtime bottlenecks on core line machinery.



Data preview — CapacityAnalysis (showing 10 of 50 points/rows)

MachineId	UtilizationPercent	CapacityId	CustomerId	ShiftId
1	100.00	1	1	4
1	51.62	2	1	1
1	52.62	3	1	4
1	100.00	4	1	3
1	100.00	5	1	2
1	61.88	6	1	2
1	67.25	7	1	1
1	59.75	8	1	3
1	64.00	9	1	4
1	76.38	10	1	1

Appendix — Data Quality

Table	Records	Completeness	Outliers	Score
cameras	3	71.4%	0	Needs review
camera_status_logs	50	100.0%	0	Good
MachineMaster	40	100.0%	0	Good
CapacityAnalysis	50	100.0%	0	Good
Machine	40	100.0%	0	Good
MachineFGMapping	50	100.0%	0	Good
MachineMaterialMapping	50	100.0%	0	Good
WeighingMachine	50	100.0%	0	Good

LLM Usage & Cost

Model	Input tokens	Output tokens	Total tokens	Estimated cost
gemini/gemini-3.5-flash	5,372	2,630	8,002	$0.0317